Question 1 : What is Power BI and why is it used in businesses?

Ans:-

Power BI is a Business Intelligence (BI) and data visualization tool developed by Microsoft. It helps organizations connect to data, analyze it, and create interactive reports and dashboards to support better decision-making.

What Power BI Does

Power BI allows users to:

- Connect to multiple data sources (Excel, SQL databases, cloud services, APIs, etc.)
- Transform and clean data using Power Query
- Create data models and calculations using DAX (Data Analysis Expressions)
- Build interactive dashboards and reports
- Share insights with teams through the Power BI Service

Why Businesses Use Power BI

Businesses use Power BI for several important reasons:

1. Data Visualization
It converts raw data into interactive charts, graphs, maps, and dashboards, making complex data easy to understand.

2. Better Decision Making
Managers and executives can analyze real-time data and make faster, data-driven decisions.

3. Integration with Many Data Sources
Power BI can connect to many platforms like Excel, SQL Server, Azure, Salesforce, and Google Analytics.

4. Real-Time Insights
Businesses can monitor live data updates, such as sales performance or operational metrics.

5. Easy Collaboration
Reports and dashboards can be shared securely with teams across departments.

Simple Example

For example, a sales company can use Power BI to:

- Track daily sales
- Compare regional performance
- Monitor profit margins
- Identify top-selling products

Question 2 : Name and explain the three main components of Power BI.

Ans:-

The three main components of Power BI developed by Microsoft are **Power BI Desktop, Power BI Service, and Power BI Mobile.** These components work together to help users create, publish, and view reports.

1. Power BI Desktop

Microsoft Power BI Desktop

Power BI Desktop is a **Windows application used to create reports and data models**.

Main Uses:

- Connect to different data sources (Excel, SQL Server, APIs, etc.)
- Transform and clean data using Power Query
- Create data models and relationships
- Write DAX formulas for calculations
- Design interactive reports and visualizations

This is mainly used by data analysts and developers.

2. Power BI Service

Microsoft Power BI Service

Power BI Service is a cloud-based platform where reports and dashboards are published and shared online.

Main Uses:

- Publish reports created in Power BI Desktop
- Create dashboards
- Share reports with teams
- Schedule data refresh
- Collaborate across the organization

Businesses use it to access reports from anywhere via browser.

3. Power BI Mobile

Microsoft Power BI

Power BI Mobile is a mobile application that allows users to view dashboards and reports on smartphones or tablets.

Main Uses:

- Access dashboards on Android or iOS
- Get real-time notifications
- Monitor business performance on the go

Mostly used by managers and executives.

Question 3 : Explain the Power BI workflow.

Ans:-

The Power BI workflow describes the process of collecting data, transforming it, creating reports, and sharing insights using tools from Microsoft. It shows how data moves from raw data to business insights.

Power BI Workflow (Step-by-Step)

1. Connect to Data Sources

Using Microsoft Power BI Desktop, you first connect to different data sources such as:

- Excel
- SQL Server
- CSV files
- Cloud services
- APIs

This step imports raw data into Power BI.

2. Data Transformation (Power Query)

After loading data, you clean and transform the data using Power Query Editor.

Common tasks:

- Remove duplicates
- Handle missing values
- Merge or append tables
- Change data types
- Filter unnecessary rows

This step prepares clean and structured data for analysis.

3. Data Modeling

Next, you create relationships between tables and build a data model.

Tasks include:

- Create table relationships
- Build calculated columns
- Create measures using DAX

This helps Power BI analyze data efficiently.

4. Create Reports & Visualizations

You then design interactive reports using charts and visuals such as:

- Bar charts
- Line charts
- Tables
- KPIs
- Maps
- Slicers

These visuals help users understand trends and patterns.

5. Publish to Power BI Service

After creating reports, you publish them to the cloud using Microsoft Power BI Service.

This allows reports to be shared with others in the organization.

6. Create Dashboards & Share Insights

In Power BI Service you can:

- Create dashboards
- Share reports with teams
- Set scheduled data refresh
- Manage user access

Reports can also be viewed on Microsoft Power BI.

Question 4 : List any four data cleaning tasks that can be performed in Power Query.

Ans:-

In Microsoft Power BI, Power Query is used to clean and transform raw data before analysis. Below are four common data cleaning tasks that can be performed in Power Query.

1. Remove Duplicates

This task removes duplicate rows from the dataset to ensure data accuracy.

Example:
If the same customer record appears twice, Power Query can keep only one record.

2. Handle Missing Values

Power Query allows you to replace or remove null/blank values.

Example:

- Replace null values with 0
- Replace missing text with “Unknown”
- Remove rows containing missing values

3. Change Data Types

You can change the data type of a column to ensure correct analysis.

Examples:

- Text → Number
- Text → Date
- Number → Decimal

This ensures calculations and filtering work correctly.

4. Split or Merge Columns

Power Query can split or combine columns.

Examples:

- Split Full Name → First Name and Last Name
- Merge City + State → Full Address

This helps organize the dataset properly

Dataset : [Real_Estate Dataset]

Question 5 : Write step by step instructions to load the above dataset.

Ans:-

Below are the step-by-step instructions to load the dataset into Power BI. The dataset contains columns like **Property_ID, Property_Name, City, Location_Details,Price, Area_sqft, Bedrooms, Bathrooms, Owner_Name, Contact**, etc.

Steps to Load the Dataset in Power BI

Step 1: Open Power BI Desktop

Open Microsoft Power BI Desktop on your computer.

Step 2: Click on "Get Data"

1. On the Home tab, click Get Data.
2. A list of data sources will appear.

Step 3: Select Data Source

1. Choose Text/CSV (since the dataset is a CSV file).
2. Click Connect.

Step 4: Browse and Select the Dataset

1. Navigate to the folder where the dataset file is stored.
2. Select the CSV file.
3. Click Open.

Step 5: Preview the Data

Power BI will show a preview window displaying the dataset columns such as:

- Property_ID
- Property_Name
- City
- Location_Details
- Price
- Area_sqft
- Bedrooms
- Bathrooms
- Owner_Name
- Contact

Check if the data looks correct.

Step 6: Load or Transform Data

You will see two options:

- **Load** → Directly load the dataset into Power BI
- **Transform Data** → Open Power Query Editor to clean or modify data

If no cleaning is needed, click Load.

Step 7: Data Loaded Successfully

After clicking Load:

- The dataset will appear in the Fields pane.
- You can start creating charts, tables, and dashboards.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

Ans:-

In Microsoft Power BI Desktop, there are three main views used to work with data and create reports: Data View, Report View, and Model View. Each view serves a different purpose in the data analysis process.

1. Data View

Data View allows you to see the data in a table format after it has been loaded into Power BI.

Purpose:

- View the dataset row by row and column by column
- Create calculated columns

- Check data values and formats
- Understand the structure of the dataset

It helps analysts inspect and validate the data before building reports.

2. Report View

Report View is where you create visualizations and build reports.

Purpose:

- Create charts such as bar charts, pie charts, line charts, and tables
- Design interactive dashboards
- Add filters and slicers
- Arrange visuals on the report canvas

This is the most commonly used view for building reports and dashboards.

3. Model View

Model View shows the relationships between different tables in the dataset.

Purpose:

- Create and manage table relationships
- Build a data model
- Understand how tables are connected
- Improve data analysis and performance

It helps ensure accurate data relationships when multiple tables are used.

Question 7 : Discuss the different data sources that Power BI supports.

Ans:-

Microsoft Power BI supports **many different data sources**, allowing users to connect, analyze, and visualize data from various platforms. These sources can be grouped into several categories.

1. File-Based Data Sources

Power BI can import data from files stored on your computer.

Examples:

- Excel files (.xlsx)
- CSV / Text files
- XML files
- JSON files
- PDF files

These are commonly used for small datasets or exported reports.

2. Database Data Sources

Power BI can connect directly to databases used by organizations.

Examples:

- SQL Server
- MySQL
- Oracle Database
- PostgreSQL
- IBM DB2

These are used when companies store large amounts of structured data.

3. Cloud-Based Data Sources

Power BI integrates with many cloud services.

Examples:

- Azure SQL Database
- Google Analytics
- Salesforce
- SharePoint Online
- Azure Data Lake

These sources allow businesses to analyze cloud data in real time.

4. Online Services / Web Data

Power BI can connect to web-based data sources.

Examples:

- Web pages (URL)
- REST APIs
- OData feeds

This is useful when pulling live data from websites or applications.

5. Other Microsoft Tools 🔗

Power BI also integrates easily with other tools from Microsoft.

Examples:

- Excel
- SharePoint
- Dynamics 365
- Azure services

This makes Power BI very powerful for companies using the Microsoft ecosystem.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Ans:-

To split the Owner_Name column into First Name and Last Name in Microsoft Power BI Desktop, you can use Power Query Editor. Follow these steps:

Steps to Split Owner_Name into First Name and Last Name

Step 1: Open Power Query Editor

1. Open your dataset in Power BI Desktop.
2. Go to the Home tab.
3. Click Transform Data to open Power Query Editor.

Step 2: Select the Column

1. In Power Query Editor, locate the Owner_Name column.
2. Click on the Owner_Name column header.

Step 3: Split the Column

1. Go to the Transform tab.
2. Click Split Column.
3. Choose By Delimiter.

Step 4: Choose the Delimiter

1. Select Space as the delimiter (because first and last names are separated by a space).
2. Choose Split at the left-most delimiter.
3. Click OK.

Step 5: Rename the Columns

Power BI will create two columns:

- Owner_Name.1
- Owner_Name.2

Rename them to:

- First_Name
- Last_Name

Step 6: Apply Changes

1. Click Close & Apply.
2. The dataset will now contain the First Name and Last Name columns.

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Group By | Use First Row as Headers | Transpose | Reverse Rows | Count Rows | Data Type: Text | Detect Data Type | Rename | Replace Values | Fill | Pivot Column | Unpivot Columns | Move | Convert to List | Split Column | Format | Merge Columns | Extract | Parse | Statistics | Standard | Scientific | Trigonometry | Rounding | Information | Date | Time | Duration | Run R script | Run Python script

Table | Any Column | Text Column | Number Column | Date & Time Column | Scripts

Queries

```
= Table.RenameColumns(#"Changed Type1",{{"Owner_Name.1", "First_Name"}, {"Owner_Name.2", "Last_Name"}})
```

	City	Location_Details	Price	Area_sqft	Bedrooms	Bathrooms	First_Name	Last_Name	Contact
1	ımbai	Andheri East - Mumbai	12500000	850	2	2	Rahul	Mehta	9876543210
2	ımbai	Borivali West - Mumbai	9800000	720	2	2	Anita	Sharma	9898989898
3	ngalore	Whitefield - Bangalore	8500000	900	3	2	Ramesh	Kumar	9123456789
4	lhi	Saket - Delhi	11000000	880	3	2	Neha	Verma	9988776655
5	ne	Hinjewadi - Pune	null	950	3	3	Amit	Joshi	9090909090
6	ngalore	Electronic City - Bangalore	7200000	800	2	2		null	9345678901
7	derabad	Gachibowli - Hyderabad	6700000	null	2	2	Suresh	Reddy	9012345678
8	ennai	Velachery - Chennai	7900000	870	3	2	Divya	Iyer	9556677889
9	ne	Baner - Pune	13500000	1200	4	3	Manoj	Patil	9887766554
10	lhi	Dwarka - Delhi	6500000	700	2	2	Raj	Singh	null
11	ımbai	Andheri East - Mumbai	12500000	850	2	2	Rahul	Mehta	9876543210
12	ımbai	Borivali West - Mumbai	9800000	720	2	2	Anita	Sharma	9898989898
13	ngalore	Whitefield - Bangalore	8500000	900	3	2	Ramesh	Kumar	9123456789
14	lhi	Saket - Delhi	11000000	880	3	2	Neha	Verma	9988776655
15	ne	Hinjewadi - Pune	null	950	3	3	Amit	Joshi	9090909090
16	ngalore	Electronic City - Bangalore	7200000	800	2	2		null	9345678901
17	derabad	Gachibowli - Hyderabad	6700000	null	2	2	Suresh	Reddy	9012345678
18	ennai	Velachery - Chennai	7900000	870	3	2	Divya	Iyer	9556677889
19	ne	Baner - Pune	13500000	1200	4	3	Manoj	Patil	9887766554
20	lhi	Dwarka - Delhi	6500000	700	2	2	Raj	Singh	null
21	ımbai	Andheri East - Mumbai	12500000	850	2	2	Rahul	Mehta	9876543210
22	ımbai	Borivali West - Mumbai	9800000	720	2	2	Anita	Sharma	9898989898
23	ngalore	Whitefield - Bangalore	8500000	900	3	2	Ramesh	Kumar	9123456789
24	lhi	Saket - Delhi	11000000	880	3	2	Neha	Verma	9988776655
25	ne	Hinjewadi - Pune	null	950	3	3	Amit	Joshi	9090909090
26	ngalore	Electronic City - Bangalore	7200000	800	2	2		null	9345678901
27	derabad	Gachibowli - Hyderabad	6700000	null	2	2	Suresh	Reddy	9012345678
28									

11 COLUMNS, 100 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 21:59

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